



ROI Training

Anthropic Models on Amazon Bedrock

Module 5:

Security, Governance, and
Guardrails

Course Contents

Module 1: Foundations: Generative AI and Claude Models on AWS

Module 2: Claude API Fundamentals and Prompt Engineering

Module 3: Data Preparation and Knowledge Bases for RAG

Module 4: Tool Use and Agentic Workflows with Claude

▶ **Module 5: Security, Governance, and Guardrails**

Module 6: Practical Architecture and Deployment Patterns

Module 7: Claude Code: Building Bedrock Applications

Module 8: Amazon Kiro and Enterprise Code Generation



Module Objectives



IAM Policies

Apply least-privilege principles for Bedrock access



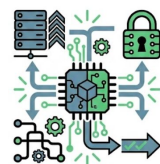
Configure Guardrails

Set up content filtering and PII protection



Monitor with CloudWatch

Track metrics, latency, and errors for Claude endpoints



Security Best Practices

Implement AWS security controls for AI workloads



ROI Training

IAM Security

AWS Shared Responsibility Model for AI



AWS Responsibility

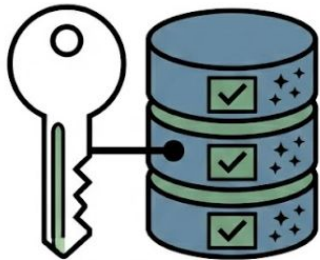
- Physical infrastructure
- Model hosting and encryption
- Network isolation



Customer Responsibility

- Data classification
- IAM and access control
- Guardrails and app audit

Amazon Bedrock Security Features



Encryption

KMS at rest

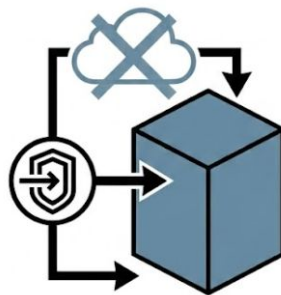
TLS 1.2+ in transit



Network

VPC endpoints

Private connectivity



Data Privacy

Never used to train models



Audit

CloudTrail logs
all API calls

IAM Best Practices for Amazon Bedrock

Least-Privilege Access Control

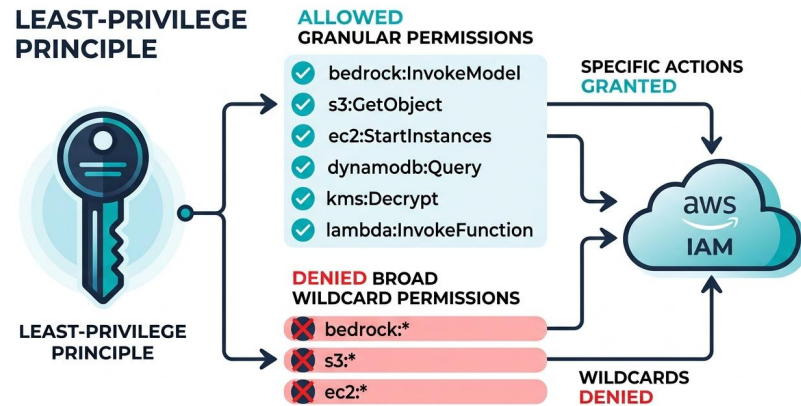
Grant only the permissions needed

Resource policies: restrict by model or KB

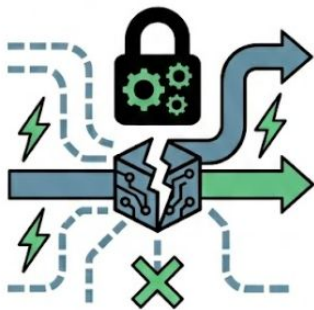
Condition keys: IP, time, MFA status

Dedicated service roles for Bedrock ops

Review and rotate permissions regularly



IAM Policy: Too Permissive vs Least Privilege

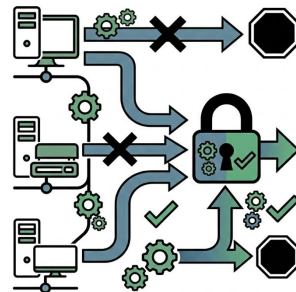


Too Permissive

bedrock:* on all resources

Wildcard = full Bedrock control

No model or region scope



Least Privilege

InvokeModel on Claude only

Resource: foundation-model ARN

Region-scoped to us-east-1

VPC Endpoints for Private Access

Private Networking

Interface VPC endpoint for
bedrock-runtime

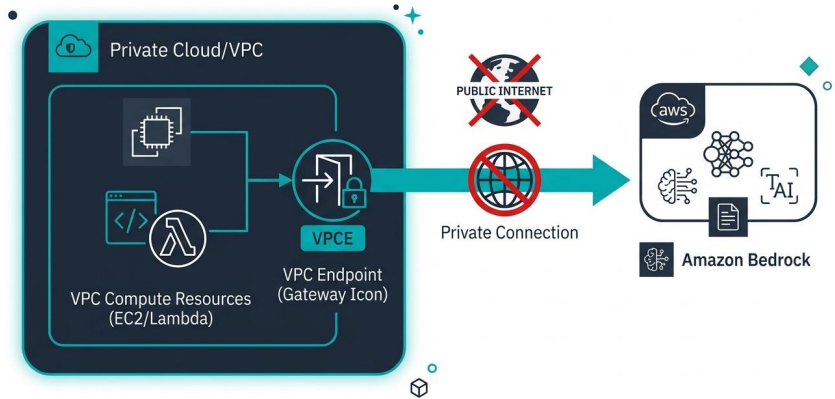
Traffic stays on AWS private network

No NAT gateway required for Bedrock calls

Apply endpoint policies for extra security

Required for many compliance frameworks

VPC Endpoints for Private AWS service access





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Guardrails

Amazon Bedrock Guardrails



Content Safety
and Governance
for Foundation
Models

Content filtering across harmful categories
Custom denied topic policies
PII detection and redaction
Automated Reasoning checks

Four Layers of Guardrail Protection

Content Filters

Block harmful categories like hate, violence, sexual

Denied Topics

Custom subject restrictions tuned to your domain

Word Filters

Direct blocking of specific terms, brands, code names

PII Detection

Mask or block SSN, email, phone, credit cards

Content Filter Categories

1

Hate: Discrimination based on identity

2

Insults: Demeaning or offensive language

3

Sexual: Sexually explicit or suggestive content

4

Violence and Misconduct: Harm or unethical behavior

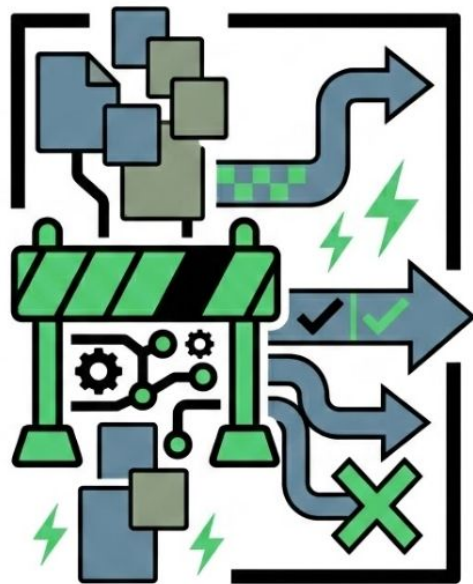
5

Prompt Attack: Attempts to bypass safeguards

Content Filter Configuration

Tunable Per Category

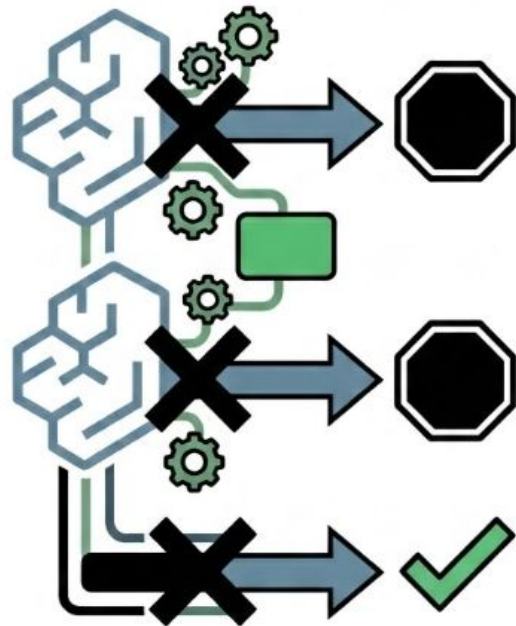
- Strength: None, Low, Medium, High
- Apply to inputs and outputs separately
- Standard tier extends to code elements
- Text and image content detection
- Tune to balance safety and false positives



Denied Topics - Custom Topic Blocking

Custom Topic Filters

- Block specific topics in queries and responses
- Define custom topics with descriptions
- Example: investment advice in banking apps
- Example: competitor product discussions
- Works on subject matter, not just tone



Word Filters - Direct Content Control

1

Custom word and phrase blocking

2

Built-in profanity filter option

3

Pattern matching support

4

Block competitor names and brands

5

Hide internal project code names

Sensitive Information Filters - PII Protection

PII Protection

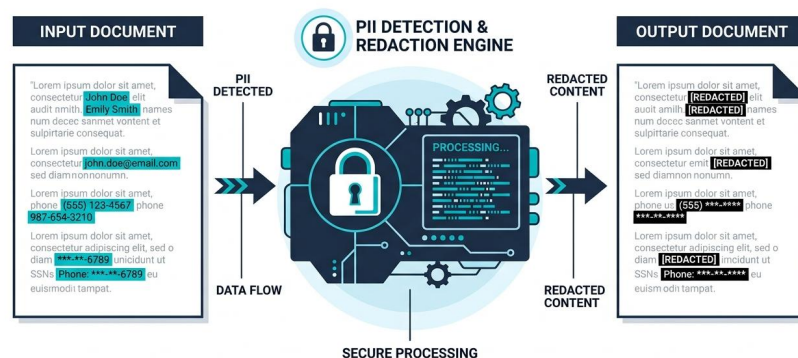
Detect SSN, DOB, email, phone, credit card

Configurable action: Block or Mask

Mask placeholders: [SSN], [EMAIL], [PHONE]

Custom regex for proprietary data formats

Supports GDPR, CCPA, HIPAA workflows



PII Detection Example

- Input: "My SSN is 123-45-6789"
- With masking: "My SSN is [SSN]"
- PII is detected but conversation continues
- Original value never reaches the model or logs

Advanced Guardrail Features

Contextual Grounding

Detect hallucinations
against sources

Automated Reasoning

Validate against
business rules

ApplyGuardrail API

Use independently
of the model

Input + Output

Filter prompts and
responses

Demo



Configure Guardrails

- Create CustomerSupportGuardrail in Bedrock console
- Configure content filters: High for Hate, Insults, Sexual, Prompt Attack
- Add denied topic: Investment Advice
- Enable PII masking: SSN, credit card, phone, email
- Test with sample input containing PII and blocked topic



ROI Training

Monitoring

Amazon CloudWatch Metrics for Amazon Bedrock

Operational Metrics

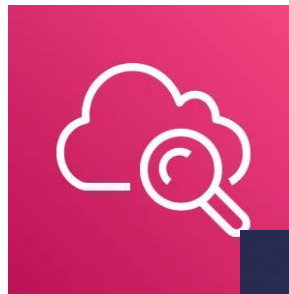
Token usage: InputTokens,
OutputTokens

Latency: ModelLatency for processing
time

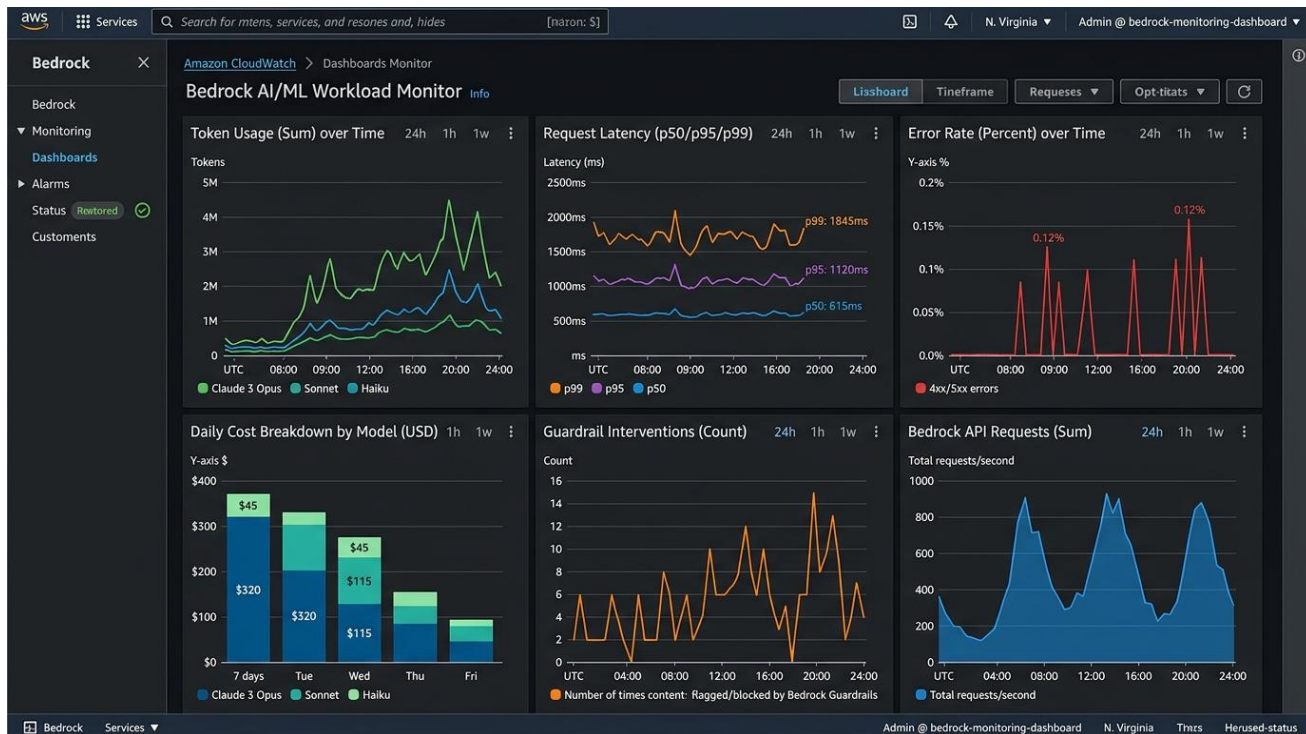
Errors: InvocationErrors,
ThrottlingErrors

Usage: Invocations,
InvocationClientErrors

Feed alarms and dashboards



CloudWatch Dashboard for AI Operations



CloudTrail for Audit Trails

1

Logs every Bedrock API call automatically

2

Who made the call (IAM identity)

3

When the call was made (timestamp)

4

What the call requested (API operation, parameters)

5

Enables compliance reporting and security forensics

Guardrail Monitoring

Monitor Guardrail Activity

GuardrailBlocked: requests fully stopped

GuardrailInterventions: requests modified

Breakdown by filter type and topic

Tune filters from real traffic data

Detect probing and abuse attempts



What We Covered



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Module 6 Is Next!

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